Improving Performance

Week 4

COMP6252 (Deep Learning Technologies)
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Introduction

- As we have already seen DNN are powerful models.
- This is mainly due to their strong expressiveness:
  - the ability to model complicated relationships between input and output
- But this same expressiveness can lead to some problems

1. Overfitting
   - Due to their power, NN can "learn" noise present in the training but not in the test datasets
   - This reduces their generalization efficacy
2. Convergence
   - Training takes long time. Sometimes doesn't even converge
   - The results highly dependent on the meta parameters

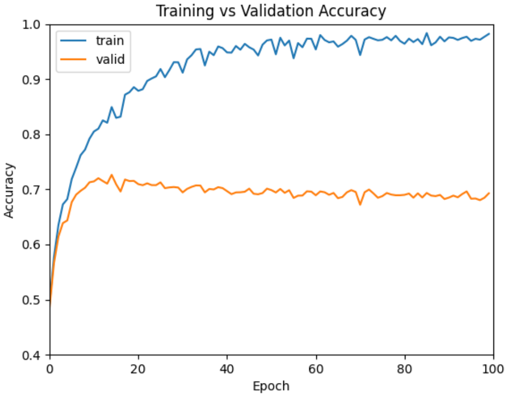
What can be done?

This week we will introduce three approaches to mitigate the problems of convergence and overfitting.

1. Early stopping
2. Dropout
3. Batch normalization
4. Data augmentation

Overfitting

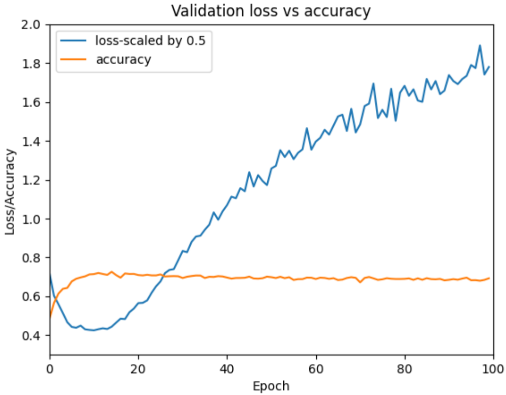
- An example of overfitting is shown below
- The training accuracy keeps increasing while the validation plateau and even decreases



- In the previous figure, the max accuracy occurs at epoch 14
- If we stop training at epoch 14 or close to it not only we get better accuracy but we save lots of time
- This is exactly what is called **early stopping**
- How do we know when to stop training?
- One of the simplest methods is to monitor the **validation loss**

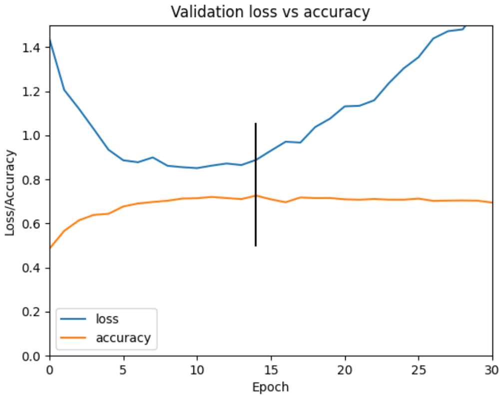
Validation loss vs accuracy

- Below is a plot of validation accuracy vs loss.
- Loss was multiplied by 0.5 to show them both on the same plot
- It is clear that accuracy plateaus after the loss starts increasing



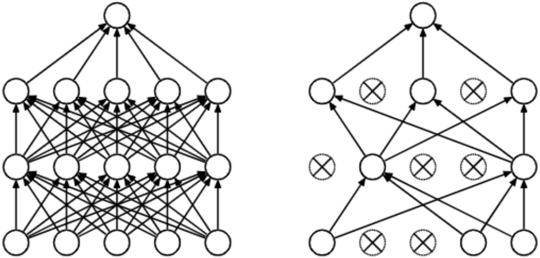
A zoomed view

- Below is a zoomed view of the previous figure
- A vertical line passing by epoch 14 (where max accuracy occurs) is shown



Dropout

- Dropout is another method used to minimize overfitting Srivastava et al. 2014.
- A dropout layer "drops" certain nodes from the NN as shown in the figure below



(a) Standard Neural Net (b) After applying dropout.

From: Srivastava et al.

What is happening exactly?

- Every training pass **different** nodes are dropped
- A neural network with n nodes that uses dropout, can be seen as a set of 2^n different "smaller" networks. Why?
- It means that using dropout is equivalent to averaging over an ensemble of Neural networks sharing the same weights
- Since the noise learned by each "smaller" network is random, "averaging" makes them cancel each other.

Implementation

- Using layers with an on/off switch on nodes is not practical
- More importantly, with an NN with large number of nodes it is **impossible**
- Instead, a node is turned off by multiplying its output with zero **with probability** p
- Mathematically, let y^l be the output of layer l in a fully connected network
- Then the output of layer $(l+1)$ is given by (* is element wise product)

$$\begin{aligned} \mathrm{r}^l &= Bernoulli(p) \\ \tilde{\mathrm{y}}^l &= \mathrm{r}^l * \mathrm{y}^l \\ \mathrm{z}^{l+1} &= \mathrm{W}^{l+1}\tilde{\mathrm{y}}^l + \mathrm{b}^{l+1} \\ \mathrm{y}^{l+1} &= \sigma(\mathrm{z}^{l+1}) \end{aligned}$$

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What about testing?

- We have seen that training a NN with dropout is is equivalent to training multiple "smaller" networks.
- The result is an average over all such networks.
- To be consistent we must use the weights learned in the training phase for testing
- This means we have to **average** over all possible networks, which is infeasible.
- We approximate this averaging by **not using dropout** during testing
- But the weights obtained in the test/validate phase are multiplied by the probability $1 - p$
- Equivalently the weights are multiplied by $\frac{1}{1-p}$ during training

Batch Normalization

- A Batch normalization layer transforms the distribution of the input to a distribution that has 0 mean and unit variance
- Batch normalization is a powerful method that
  1. Allows for better convergence of NN training
  2. Adding BN layers to existing NN yield better generalization results
  3. Reduces over fitting
  4. Training is less sensitive to meta-parameters (learning rate, weights initialization)

Batch Normalization- How does it work

- Given a mini-batch of tensors x_{ci} of dimension (S,C,H,W)
- where c is the channel index and i collectively refers to all other dimensions.
- Let $N = S \times H \times W$.
- Batch normalization computes the mean and variance of the batch (per channel) according to

$$\mu_c = \frac{1}{N}\sum_{i=1}^{N} x_{ci}$$

$$\sigma_c^2 = \frac{1}{N}\sum_{i=1}^{N} (x_{ci} - \mu_c)^2$$

Given input x_{ci} the normalized inputs of the layer are computed as follows:

$$\hat{x}_{ci} = \frac{x_{ic} - \mu_c}{\sqrt{\sigma_c + \epsilon}}$$

Where ϵ is a small number added for numerical stability. Therefore, for each channel, the $\hat{x}_{ci}$ have zero mean and unit variance.
The output of the batch normalization layer is given by

$$y_{ic} = \gamma_c \hat{x}_{ic} + \beta_c$$

Where γ_c and β_c are **learnable** parameters.

Batch normalisation: inference

- At every step during training, μ_c and σ_c are computed using the mini-batch as described above
- Also, running averages of the mean and variance $\hat{\mu}_c$, $\hat{\sigma}_c$ are computed
- Specifically, at each step when μ_c and σ_c are computed, and the running averages are updated as follows:

$$\hat{\mu}_c \leftarrow 0.9\hat{\mu}_c + 0.1\mu_c$$
$$\hat{\sigma}_c \leftarrow 0.9\hat{\sigma}_c + 0.1\sigma_c$$

- During inference, for input x_{ci}, the output of the batch normalization layer is given by

$$y_{ic} = \frac{x_{ic} - \hat{\mu}_c}{\sqrt{\hat{\sigma}_c}}$$

Batch normalization - Why does it work?

- Considerable ongoing debate
- In the original paper of Ioffe and Szegedy 2015 the claim is that it fixes the internal covariate shift
- meaning the change in the distribution due to the change of the weights during learning
- A different interpretation was given by Kohler et al. 2019: it separates the change in magnitude from the change of directions.
- This can be seen from the fact that the $\hat{x}_{ci}$ are normalized and all of them share the same magnitude γ
- Another recent interpretation can be found in Chai et al. 2020

Example

Consider a mini-batch of size two, containing the tensors of size 2×2, A and B with both having a single channel.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

- The mean is 4.5 and (biased) variance is 5.25 therefore the output is

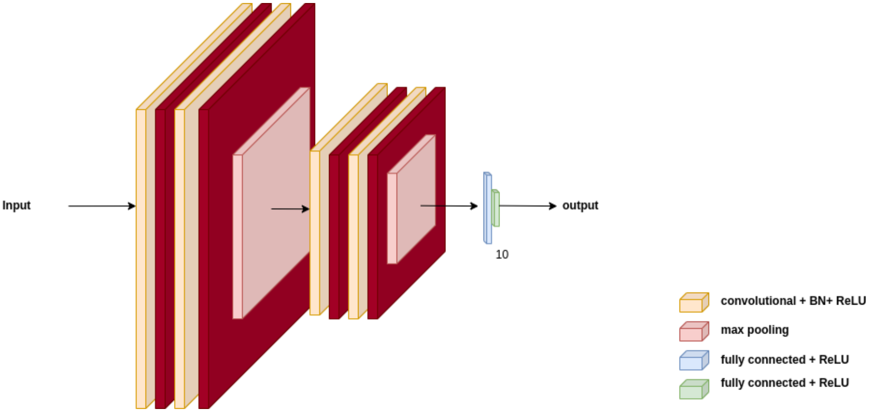
$$A = \begin{bmatrix} (1-4.5)/5.25 & (2-4.5)/5.25 \\ (3-4.5)/5.25 & (4-4.5)/5.25 \end{bmatrix} \qquad B = \begin{bmatrix} (5-4.5)/5.25 & (6-4.5)/5.25 \\ (7-4.5)/5.25 & (8-4.5)/5.25 \end{bmatrix}$$

- When the input has multiple channels, the same operation is performed for each channel independently

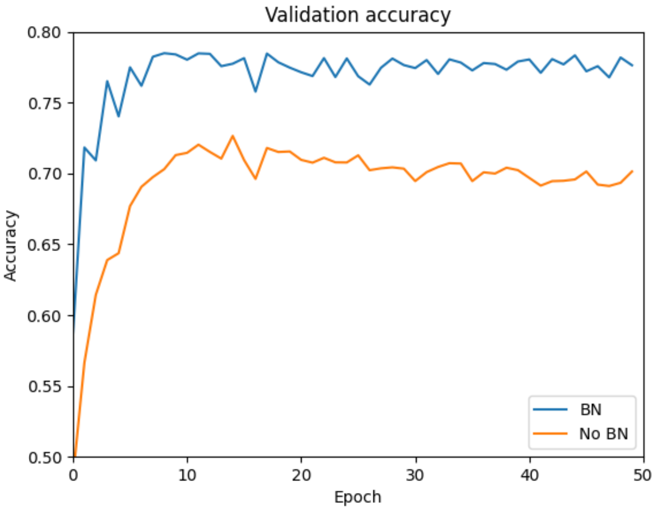
In practice

- Data: CIFAR10
- The NN has two convolutional blocks
- The first contains two sub-blocks each with
  1. 32 filters with receptive field of 3x3
  2. followed by BN layer
  3. followed by ReLU
- the sub-blocks are followed by a max pooling layer of size 2x2
- The second block is the same as the first but with 64 filter
- The conv blocks are followed by two feedforward layers with 128 and 10 nodes respectively

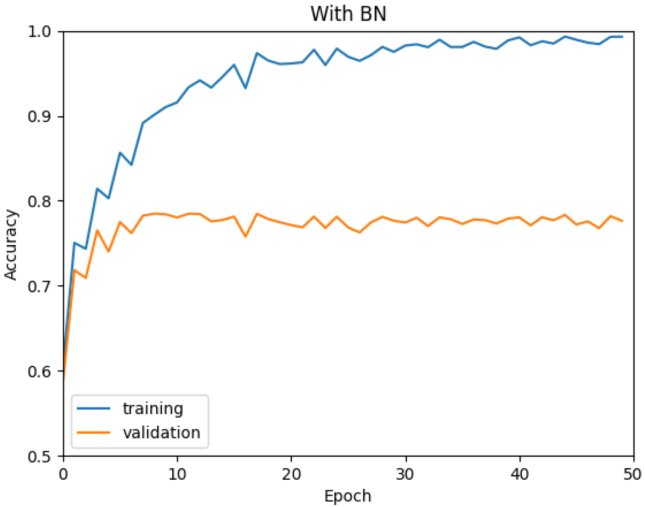
Nework Architecture



Results: validation accuracy of BN vs no BN

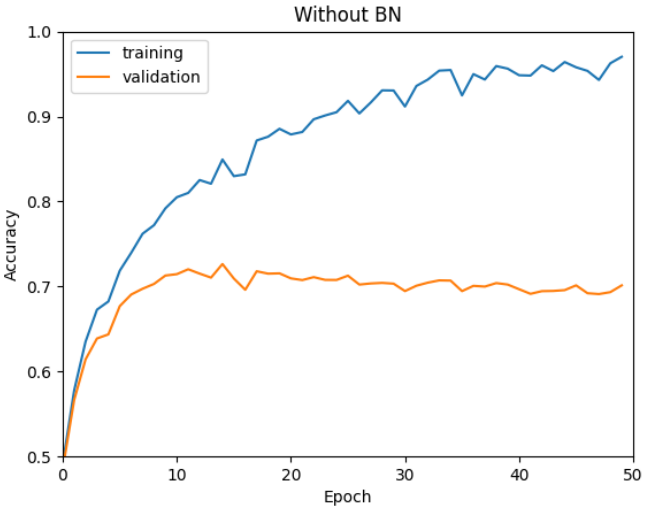


Results: train-vs-validation accuracy with BN



- Batch normalization give much better accuracy.
- Peak value for the accuracy for BN is 78% vs 72%
- The average after the peak for BN is vs 75% vs 69%
- Also, with BN the peak acc is reached after 8 epochs vs 14 without BN.

Results: train-vs-validation accuracy without BN



Data augmentation

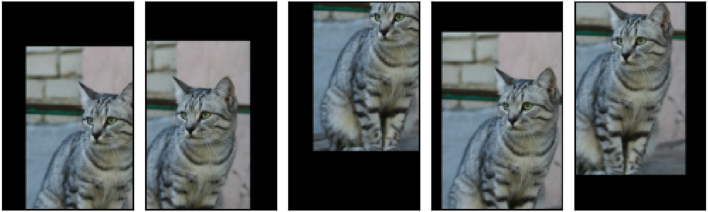
- Data augmentation is one method of obtaining higher test/validation accuracy
- Data augmentation is typically done "on-the-fly" in PyTorch
- This is done by applying some transforms to the input
- Example transforms
- Rotation
- Translation
- Horizontal flip
- Random crop
- ......

Example: RandomHorizontalFlip

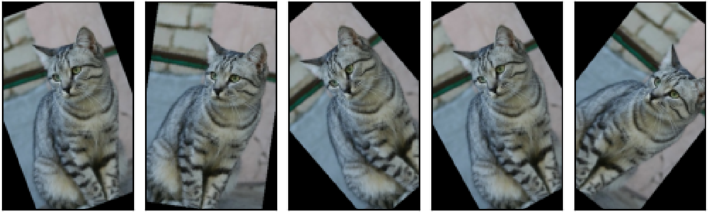
- Horizontally flip the image with probability 0.5
- RandomHorizontalFlip(p=0.5)

Example: Translation

- RandomAffine(degrees=0,translate=(0.3,0.3))
- Used here for translation only
- Can also perform *rotation*, *scale*, *shear*,etc...



- RandomRotation(45)
- Rotate between -45 and 45 degrees



Example: Random crop

```
RandomCrop((150,150))
```

Auto augmentation methods

- AutoAg methods use a policy to transform the input
- For example which transform to use (e.g. rotation,crop) with what probability
- What values are used for the parameters (e.g. angle)
- Some of use search algorithms to determine the policy (e.g. AutoAugment, Cubuk et al. 2018)
- Some are very simple (e.g. TrivialAugment, Müller and Hutter 2021)

SGD with momentum

- A simple variant of SGD to reduce fast changing gradient
- It usually leads to better training convergence
- Let w be the learnable weights and α the learning rate
- Instead of

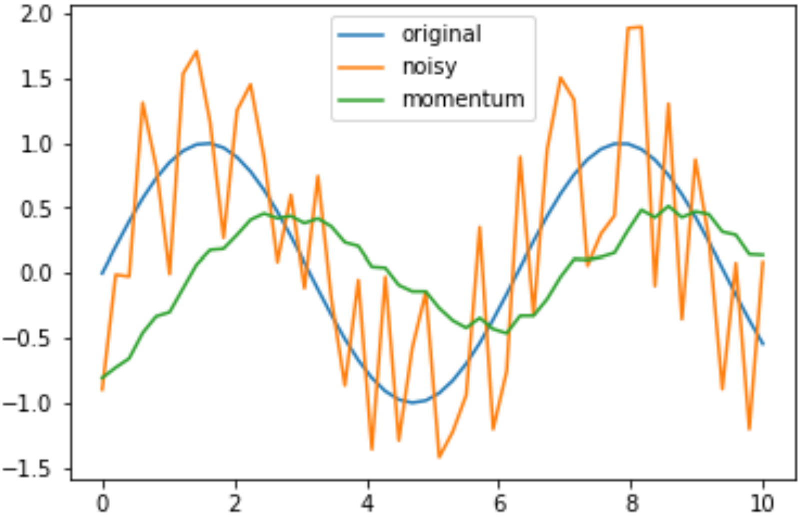
$$w = w - \alpha \nabla_w \mathcal{L}$$

- use

$$\begin{aligned} dw &= \beta dw + (1-\beta)\nabla_w \mathcal{L} \\ w &= w - \alpha dw \end{aligned}$$

- where β is a parameter, usually 0.9

Why is it called moving average?



Why is it called exponential moving average?

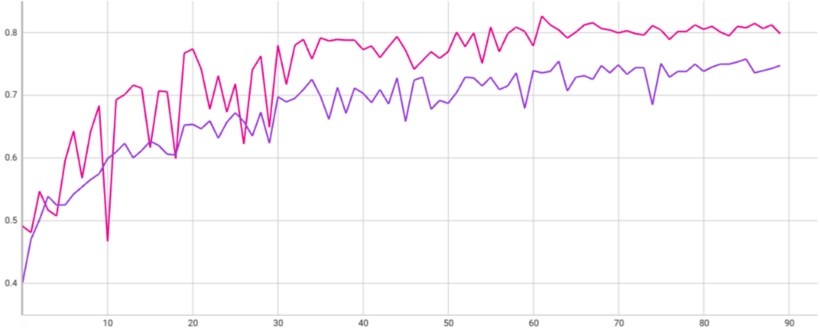
$$
\begin{aligned}
dw_t &= \beta dw_{t-1} + (1-\beta)\nabla_t\mathcal{L} \\
&= \beta(\beta dw_{t-2} + (1-\beta)\nabla_{t-1}\mathcal{L}) + (1-\beta)\nabla_t\mathcal{L} \\
&= \beta^3 dw_{t-3} + \beta^2(1-\beta)\nabla_{t-2}\mathcal{L} + \beta(1-\beta)\nabla_{t-1}\mathcal{L} + (1-\beta)\nabla_t\mathcal{L} \\
&= \beta^{t-1}(1-\beta)\nabla_1\mathcal{L} + \beta^{t-2}(1-\beta)\nabla_2\mathcal{L} + \ldots + (1-\beta)\nabla_t\mathcal{L}
\end{aligned}
$$

- Usually $\beta < 1$ so $a > b \Rightarrow \beta^a < \beta^b$
- Moreover, $n \to \infty \Rightarrow \beta^n \to 0$
- and $\beta = 0 \Rightarrow dw_t = \nabla_t\mathcal{L}$

SGD with momentum

- Validation accuracy vs epoch for ResNet18 with random initial weights

1. SGD with momentum 0.9: pink
2. Vanilla SGD: purple



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